

Nominal Rigidities in Online Prices: An Analysis of Differences Across Types of Online Businesses

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Abstract: This study analyzes price rigidity in online markets across different types of businesses. Using high-frequency microdata collected through web scraping in Colombia, it aims to complement and update price adjustment statistics for the region, identifying differences among three types of online markets: multichannel supermarkets, discount stores, and price comparison websites. The results highlight the importance of considering the type of business when estimating price rigidity in such studies. Specifically, the implicit duration of prices in the region's online markets ranges from 3 to 9 weeks, with adjustment sizes varying between 0.28% and 23%, confirming a greater price adjustment capacity in online markets compared to physical markets.

Resumen: Este estudio analiza la rigidez de precios en mercados en línea considerando distintos tipos de negocios. Utilizando microdatos de alta frecuencia recolectados mediante *web scraping* en Colombia, busca complementar y actualizar los estadísticos sobre el ajuste de precios en la región, identificando diferencias entre tres tipos de mercados en línea: supermercados multicanal, tiendas de descuento y páginas de comparación de precios. Los resultados resaltan la importancia de considerar el tipo de negocio al estimar la rigidez de precios en este tipo de estudios. Específicamente, se encuentra que la duración implícita de los precios en los mercados en línea de la región varía entre 3 y 9 semanas, con tamaños de ajuste que oscilan entre el 0.28% y el 23%, lo que confirma una mayor capacidad de ajuste de precios en los mercados en línea en comparación con los mercados físicos.

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1. Introduction

One of the most important issues for macroeconomists is whether nominal rigidities affect the real economy (Eichenbaum, Jaimovich, & Rebelo, 2011). Nominal rigidities involve slower price adjustments in response to economic shocks, which explains much of the impact of demand shocks on output, particularly the effects of monetary policy (Nakamura & Steinsson, 2013). According to standard macroeconomic models, such as New-Keynesian models, the impact of monetary policy on output in the absence of nominal rigidities is null, as prices adjust immediately in response to monetary shocks. In contrast, the slower the price adjustment, the greater the impact of a monetary shock on production.

Technological advancements have enabled firms to adjust prices more swiftly in situations where, previously, the costs associated with price changes (menu costs) rendered them economically unfeasible. In the past, these costs reduced the frequency of price adjustments. However, with the digitalization of business operations and the automation of processes, retailers can now change the prices of their products quickly and at a low cost (Kauffman & Lee, 2010). This flexibility is especially relevant in online sales channels, where prices can be adjusted almost in real time in response to fluctuations in demand, competition, or even monetary policies.

Although price rigidities have been widely studied in recent years due to the availability of new databases for researchers, these estimates may present certain issues. First, such data may require imputing missing values, have a low frequency of collection, and necessitate

aggregation, as they are not collected with the purpose of measuring price adjustment frequency (Cavallo, 2018). Second, the recent adoption of technology has changed how prices are adjusted, especially in online markets. In this new business environment, the presence of algorithms that facilitate automated price changes and other technologies has transformed the costs associated with price changes (menu costs), market monitoring costs, and search costs faced by consumers (Gorodnichenko & Talavera, 2017; Brynjolfsson, Dick, & Smith, 2010). Finally, the growing participation of e-commerce in various sectors of the economy, which has expanded to include all types of goods, makes it increasingly relevant to study online markets to understand their differences from physical markets.

Currently, there are some studies that estimate price adjustments for online markets (Gorodnichenko, Sheremirov, & Talavera, 2018; Gorodnichenko & Talavera, 2017; Cavallo, 2018; Lünemann & Wint, 2011; Ellison, Snyder, & Zhang, 2018; Ellison & Ellison, 2009). However, these studies often rely on data from a single company, typically price comparison websites or large multichannel supermarkets, which may lead to differing estimates depending on the type of business. In this context, this research aims to contribute to the understanding of price adjustments, particularly in online markets, by demonstrating that significant differences exist in price adjustment estimates depending on the business model of the company. These differences have important implications for monetary policy. As one of the first studies to simultaneously analyze different online channels within the same region, it seeks to provide a broader estimation of price adjustments for this type of market.

Using a daily database for four companies in Colombia, classified into three distinct business types, this study aims to gain a more precise understanding of the mechanisms behind price adjustments in online markets, particularly by highlighting significant differences across

various business types. The sample includes an emerging delivery company that enables the simultaneous comparison of prices across multiple businesses and is classified as a Price Comparison Website (PWCP). Additionally, a Multichannel Discount Store (TDM) and two Multichannel Large Supermarkets (GSM) are analyzed, both of which primarily operate in physical markets. Since each type of business follows a distinct operational model, significant differences in their price adjustment mechanisms are expected. These differences, often overlooked when analyzing data from a single type of business, are crucial for a comprehensive understanding of price behavior in online markets.

While a higher frequency of price adjustment is observed compared to physical markets—a result commonly found in the literature—the differences between business types can, in some cases, be up to three times larger than in others. This variation is not limited to adjustment frequency; significant differences are also found in adjustment sizes, which can be up to 20 times smaller in discount stores. These findings contribute to a deeper understanding of the price adjustment mechanism in online markets, highlighting the importance of complementing estimates with multiple data sources.

Consumption patterns between physical and virtual markets have undergone a significant transformation since the creation of the internet, a phenomenon that was exacerbated by the pandemic. Globally, more and more people are choosing to make their purchases through virtual channels. Currently, e-commerce accounts for 20.1% of total global retail sales (Gelder, 2024). The average monthly share of e-commerce in total retail sales for Colombia in 2023 was 2.2%, according to data from the Cámara de Comercio Electrónico (2024b). Although this share is still modest, this type of commerce is expected to continue developing, with a projected growth of 17% for 2024 (Cámara de Comercio Electrónico, 2024b).

According to the Colombian Chamber of Electronic Commerce (2024a), 73% of consumers report buying more through these channels than before the pandemic, with virtual sales reaching 62.1 trillion Colombian pesos (COP) in 2023. This represents a 12.58% increase compared to the previous year in nominal terms. Considering the annual inflation rate for 2023, which was 9.28% according to the National Department of Statistics (2024), the real growth exceeded 3%.

Technology, other retail¹ and entertainment are the most consumed categories within the country's e-commerce (accounting for 27.6%, 15.2%, and 10.9%, respectively). In particular, the food delivery services sector is experiencing rapid growth due to increased technological adoption and the convenience it offers consumers (Statista Market Insight, 2024). Given the minimal menu costs, it becomes relevant to complement the estimates with high-frequency data that allow identifying patterns consistent with the new price adjustment capacity of these firms. To this end, this study uses daily data downloaded directly from the websites of three types of businesses distributed across four online stores, which will help provide a broader understanding of price adjustments in the online market. By including different types of businesses, it aims to propose a wider and more precise range for estimating the frequency, size, and probability of price changes.

In summary, virtual channels are becoming increasingly important in the global economy, highlighting the relevance of complementing traditional studies with new sources of information obtained from the web. These channels often allow for more frequent price adjustments than physical stores, which could potentially limit central banks' ability to influence economic activity. For data collection, the technique of web scraping is used, which

¹ According to the Colombian Chamber of Electronic Commerce (2024a), this category includes retail, food, stores, restaurants, and others.

allows downloading all available information from a website and storing it in databases. Studies have shown the advantages these data sources can offer, especially due to their low cost and high frequency of collection (Cavallo, 2017; Hillen, 2019). Their fields of application are varied, including business intelligence and computer science (Khder, 2021), inflation measurement (Cavallo, 2017; Benedetti, Laureti, Palumbo, & Rose, 2022), the labor market (Schedlbauer, Raptis, & Ludwig, 2021), the housing market (Boeing & Waddell, 2017), and food prices (Hillen, 2019).²

Price rigidities have been primarily studied using two different data sources: directly scanned from physical stores and microdata from the consumer and producer price indices (Eichenbaum, Jaimovich, & Rebelo, 2011; Nakamura & Steinsson, 2008; Nakamura & Steinsson, 2013; Julio & Zárate, 2008; Bils & Klenow, 2004; Álvarez, et al., 2006). Although these types of data have proven useful for documenting the characteristics of price adjustments, there are several reasons to exercise caution. As Cavallo (2018) shows, the data commonly used in price adjustment estimates have undesirable characteristics, such as low frequency, imputations, and aggregations. These not only ignore potential patterns in higher-frequency data but also introduce biases in the frequency, size, and likelihood of price adjustments.

Currently, other studies have documented the main statistics of online markets, finding significant differences compared to physical markets. However, few studies have focused on the differences within virtual markets, which can lead to different estimates depending on the type of company. Using data from a price comparison website, Gorodnichenko & Talavera (2017) highlight how the speed of price adjustments can be affected by the competitive nature

² Refer to Cavallo (2018), Cavallo & Rigobon (2016), Cavallo (2017), Benedetti, Laureti, Palumbo, & Rose (2022) o Schedlbauer, Raptis, & Ludwig (2021) for more details about the characteristics of this type of data.

of the market, which tends to be higher in these webpages. In this regard, Ellison & Ellison (2009) highlight the importance of obfuscation by online stores, the differentiation of sellers through reputation, and the low search costs faced by consumers as key factors in the competition on such websites. Ellison, Snyder, & Zhang (2018), on the other hand, emphasize the importance of administrative costs as a source of price rigidity in these types of markets. Another important feature of these studies is that they mostly focus on the characteristics of durable goods, such as DVDs, books, laptops, and other goods that were typically purchased on websites in the past. However, it is now more common to trade all types of goods.

On the other hand, studies such as Cavallo (2018) complement price rigidity estimates by using web-scraped data from large multichannel supermarkets across different countries. This study concludes that new web-scraping data sources could improve traditional price adjustment estimates, as it shows how aggregation and imputation methods yield higher adjustment frequency estimates when using scanner data or CPI microdata—sources predominantly used in these types of studies. As highlighted by Ellison & Ellison (2009), price adjustment statistics for large multichannel supermarkets tend to be less frequent and of greater magnitude compared to price comparison websites, partly due to their less competitive environment. This underscores the importance of considering the type of business when analyzing price adjustment statistics. Instead, this study uses data from three different types of online markets—multichannel supermarkets, discount stores, and a price comparison website—arguing that these can have significant implications for price adjustment statistics due to the high variability observed among them.

In Colombia and Latin America, price rigidities have been explored using various data sources, such as microdata used in the calculation of the CPI and PPI, data provided by

retailers, national information systems, and more recently, data obtained from the web (Julio, Zárate, & Hernández, 2010; Giuliatti, Otero, & Waterson, 2020; Núñez, Otero, & Trujillo-Barrera, 2023; Julio & Zárate, 2008; Cavallo, 2018; Coronado, Lahura, & Vega, 2020). Regarding this literature, this study aims to contribute in several ways. First, it is one of the pioneering studies in analyzing online markets and the first to examine the differences between different sellers in these markets. Second, it expands the literature on pricing by understanding how businesses' behaviors differ in different web environments. Finally, it is the first to study a price comparison website in the region, which not only helps understand the characteristics of price adjustments in online markets but also distinguishes the differences between sellers.

Using high-frequency microdata on online prices in Colombia, this study finds significant differences in price adjustment patterns across business types. Discount stores exhibit longer implicit price durations and smaller price changes than their counterparts. Contrary to previous findings, price comparison websites adjust prices less frequently than multichannel supermarkets, though with similar magnitudes—potentially due to competition with large supermarkets and high monitoring costs. Overall, price adjustment statistics vary notably, with implicit durations ranging from approximately 3 to over 9 weeks and adjustment sizes between 0.28% and 23% of the previous price.

This document is structured as follows: it begins with the present introduction, which outlines the motivation for the work and reviews the literature on price rigidities. Section 2 describes the main characteristics of the data, the methodology for its collection, and the key statistics of the sellers. In Section 3, the methodology used to analyze price adjustment is defined and explained. Section 4 presents the results of the study, including the estimation of the most

relevant statistics for price adjustments and the differences observed between online businesses. Finally, Section 5 concludes with a summary of the main findings and their implications in terms of economic policy.

2. Data

This study is based on a dataset comprising 819,281 observations of 14,425 products. The information was collected from four sources: the website of two major multichannel supermarkets, which primarily operate through physical stores; a multichannel discount store following a hard discount model, characterized by offering very low prices on a limited selection of products and with a strong focus on physical stores; and a price comparison website belonging to a delivery company. The selection of these stores was based on their relevance in the local market and their presence in online commerce, ensuring the inclusion of the main players in the Colombian market according to their size and market share.

For the analysis, this sample was distributed across different periods based on data availability for each retailer. The primary analysis was conducted using a sample collected between April and July 19, 2024, which contained simultaneous information for all four retailers. The sample for the second analysis includes the entire data collection period, from August 7, 2023, to July 18, 2024. However, this dataset includes periods where data for one or more retailers were unavailable.

The products mainly consist of non-durable consumer goods, including food, non-alcoholic beverages, alcoholic beverages, and tobacco. Additionally, there are some durable consumer goods, primarily household appliances, home essentials, healthcare products, clothing, and transportation-related items. The dataset includes fields collected from the websites of each

business, such as product description, brand, price, sale price, discount, date, and the search link for the products. For the delivery platform, the dataset also includes the store, as this platform aggregates offers from different retailers.

The data collection was automated through a web scraping algorithm that gathers all available information from a specific URL. This technique allows storing all the information from the webpage available at a given URL in HyperText Markup Language (HTML). From there, the information must be structured into databases using regular expressions, which are responsible for finding text patterns in the HTML. This operation was performed daily by scheduling a task on a server at ICESI University in Cali, Colombia. The Python programming language was used to develop the code. As a result, a sample of nearly 1 million day-store-product data points was obtained, collected from three of the five largest retail businesses in Colombia and one of the main food delivery applications in the country.³

Table 1 presents the distribution of data across retailers and the main price statistics used in this study. The primary analysis focuses on the differences between various online sellers, so a subsample of the full dataset is used, ensuring that information is available for all sellers during the same period. The number of products varies across retailers, with A1 and A2 offering the largest selection (3,769 and 3,355 products, respectively), followed by B (1,339), and finally, the discount store C (327). Similarly, prices tend to be highest at the multichannel supermarkets (A1 and A2), followed by the price comparison website (B), and lowest at the discount store (C).

Table 1

Data distribution

Seller	Sample	Days	Products	N	Precio			Log(Precio)		
					Mean	SD	Median	Mean	SD	Median
C	Sub	60	327	18.705	\$7,864.93	\$8,407.51	\$5,490.00	\$8.63	\$0.80	\$8.61
	Full	307	359	36.278	\$7,522.92	\$8,204.86	\$5,000.00	\$8.57	\$0.82	\$8.52
	Sub	61	3,769	141.570	\$95,886.25	\$451,101.95	\$29,700.00	\$10.3	\$1.38	\$10.30

To conduct a robustness analysis of the statistics obtained in the main analysis, all available store prices were incorporated without restricting the timeframe. This approach aims to support the findings related to the frequency, duration, and magnitude of price adjustments, particularly since the adjustment measures in the main analysis were limited to a maximum period of approximately 60 days. Due to this restriction, measures sensitive to extreme values, such as the mean, were affected. To address this issue, the full set of collected data was used, which is heterogeneously distributed across different sellers.

Table 1 also presents the main statistics of this dataset, summarized in the "Total" row. Notably, the multichannel large supermarkets and the discount store have a greater number of observed days (over 300 days) compared to the price comparison website, which has a more limited observation window of 83 days. Despite these differences in the observation period, product price statistics remain relatively consistent: the lowest prices are found in the discount store, followed by the price comparison website, while the large multichannel report the highest prices.

Additionally, in both analyses, two types of filters were applied to ensure the presence of observations for each product throughout the entire time window. For each period, the first criterion considered was the number of observations per month, which had to be at least four. Secondly, products that met this condition were further filtered to include only those present in every month of the sample. That is, products without at least four observations in each month of the sample were removed.

3. Methodology

To understand how quickly prices adjust, several studies have focused on the frequency with which they change (Eichenbaum, Jaimovich, & Rebelo, 2011). However, one aspect highlighted in the literature is that different price adjustments can have substantially different macroeconomic implications (Kehoe & Midrigan, 2007; Klenow, 2010). There is an ongoing debate about how promotion prices should be defined and how to handle them. Some authors argue that promotional prices respond to idiosyncratic shocks, which results in a smaller aggregate impact on the price level (Nakamura & Steinsson, 2008; Eichenbaum, Jaimovich, & Rebelo, 2011; Ray, Snir, & Levy, 2023). For this reason, it is common in price adjustment literature to use different series, aiming to observe how estimates of the frequency of price adjustment can be affected by various methodologies for classifying temporary prices.

For the above reason, it is essential to differentiate between transitory and permanent price changes. This differentiation is particularly difficult in the context of this study, as compared to microdata used for constructing price indices, the data extracted from the web lacks information about this variable. To address this, two methodologies proposed by Nakamura & Steinsson (2008) and Eichenbaum, Jaimovich, & Rebelo (2011) are employed to remove temporal patterns in prices, aiming to identify price stickiness more broadly. Following these studies, new price series are constructed for each product, and the price rigidity or inertia they exhibit is analyzed.

3.1. Regular Price

The regular price is the highest price a consumer faces within a time window. This methodology removes temporary markdown prices by applying a series of conditions on the past and future prices of the product series. Using a six-step algorithm, Nakamura & Steinsson (2008) propose the following "promotion filter" to identify "V Patterns":

1. If $p_t < p_{reg}$ then $p_{reg} = p_t$.
2. If $p_t > p_{reg}$ then $p_{reg} = p_t$.
3. If $p_t > p_{reg}$ and the price never goes above p_{reg} before returning to p_{reg} then $p_{reg} = p_t$.
4. If the set S has k or more different elements, then $p_{reg} = p_t$.
5. Define p_{reg} and p_t if $p_t > p_{reg}$, then $p_{reg} = p_t$.
6. .

Where p_t is the price and p_{reg} is the regular price for the product at time t . For the first period ($t=1$) of the series, the algorithm begins at step three. It has three parameters: k, α, β , which affect how the regular price is defined. Specifically, the parameter k affects the number of future prices observed to define the regular price in the current period, while α and β affect the number of different future prices considered to define the regular price. Due to the characteristics of the data, this study employed $k=1$ to consider one and two weeks ahead.

3.2. Reference price

Following Eichenbaum et al. (2011), a reference price is defined as the most frequent price or mode within a time window. This approach helps capture a type of rigidity not observed through other strategies for identifying temporary prices, such as V-filters. As Eichenbaum et al. (2011) note, nominal rigidities can take the form of inertia in reference prices, reflecting greater price rigidity. The authors argue that, although prices may change with some frequency, they fluctuate around a reference value that remains constant for a longer period.

This methodology offers an advantage when working with high-frequency prices, as it allows for easy identification of promotional prices and addresses the issue of missing values by implicitly observing the regular price during these days. However, it is true that the results can

vary considerably depending on the time window considered. For this reason, two-time windows are employed in this study: one weekly (main case) and another monthly (annex), to capture the heterogeneity that may exist in price adjustments.

As an example of previous definitions, consider a price series for a hypothetical product, with the following observations in a given week: {\$2, \$3, \$2, nan, \$2, \$1, \$2}. The reference price of this series assuming a weekly range is \$2, as it is the most frequently observed price within this time window (observed 57% of the time within the time window), constituting the following regular price series: {\$2, \$2, \$2, \$2, \$2, \$2, \$2}. On the other hand, using the filter proposed by Nakamura (2008), the following regular price series is obtained: {\$2, \$3, \$2, nan, \$2, \$2, \$2}. As can be seen, there are differences between both methodologies. In the first, it is possible to latently define the regular price for those observations where the price record is missing, whereas in the second, this is not possible. Another difference is that the number of distinct regular prices is limited to the time window used in the first methodology, since the regular price will always be equal to the mode during the time window.

By identifying regular and reference price series, it is possible to detect temporary prices and, consequently, classify price changes. Permanent changes refer to those shifting toward a regular price, while transitory changes correspond to those moving toward a temporary price. This approach aims to incorporate a price promotion identifier, similar to those found in other datasets. Prices associated with promotions are highly transitory and typically revert quickly to their previous regular price (Nakamura, 2008). For this reason, some authors, such as Kehoe & Midrigan (2007), argue that the temporary nature of price adjustments linked to promotions implies that the same number of price changes due to these factors results in a significantly smaller impact compared to an equivalent number of changes in regular prices.

To this end, several estimates of price adjustments are made: the baseline estimate, which directly considers the posted prices in the stores, the estimate for regular prices, and the estimate for reference prices. These estimates aim to generate different measures of price rigidity, which will allow for the identification of different effects for the same monetary policy.

3.3. Adjustment frequency

The frequency of price adjustment is defined as the number of times a product's price experienced a positive absolute variation divided by the total number of possible variations within a given period. Let t be the number of days that have passed since the last time the price for product i in the store s was last observed. The variation indicator function is defined as I_t with $t > 1$. Additionally, let I_t with $t > 1$, the indicator function for a non-zero price change. The Price adjustment frequency for product i in store s is then defined as:

Using this definition and under the assumption of constant risk, the implied duration of the price in days is given by .

3.4. Adjustment Size

The adjustment size is defined as the average value of non-zero absolute variations for each product. Using the previous definitions, the adjustment size for product i in the store s is defined as:

3.5. Direction of adjustment

The direction of adjustment reflects the proportion of price changes that resulted in a price increase. Let I_{it} be the indicator function for a price increase for product i in store s at time t . Using the previous notation, the direction of adjustment, expressed as the percentage of price increases over the total number of changes, is defined as:

3.6. Transition matrix.

Let A be the transition matrix, where its entries represent the probability of moving to a specific state, conditional on the state in the previous period. Specifically, the entry a_{11} denotes the probability that a price remains in a regular state, given it was in this state the previous day. Conversely, a_{12} presents the probability that a price transitions to a temporary state, conditional on being regular the previous day. Thus, a_{21} defines the probability that a price transitions to a regular state, given that it was in a temporary state the previous day. Lastly, the entry a_{22} represents the probability that a price remains in a temporary state, given that it was also temporary the previous day.

To calculate probabilities, the number of price changes from one state to another is counted and divided by the total possible price changes. Let I_{ijt} be the indicator function for a change from state i to state j , where $\mathbf{p}_i$ and $\mathbf{p}_j$ are the vectors of all possible prices for states i and j , respectively. Using the previous notation and that from Section 3.3., the probability of transitioning from state i to state j is defined as:

4. Results

4.1. Main Analysis

For the main analysis, a 7-day window was used to filter regular prices and a one-week window to define the mode for reference prices. Additionally, the annexes include statistics obtained using 14 days and one month to define regular and reference prices, respectively.

4.1.1. Frequency of adjustment and Implied Duration

Once the adjustment frequencies and implied durations in days are calculated for each of the three product price series (posted prices, regular prices, and reference prices), the mean and median of the products by store are computed to obtain the aggregate value. These values are reported in Table 3. The adjustment frequencies for posted prices range from 0% to 6% when using the median as the aggregation method and from 1% to 11% when using the mean of the products. Similarly, the implied duration for this group ranges from 18 to 59 days when considering the median and from 27 to 58 days when considering the mean.

Table 3.

Frequency of adjustment and Implied Duration

Seller	Posted				Regular				Reference			
	Median		Mean		Median		Mean		Median		Mean	
	F (%)	(Day s)	F (%)	(Day s)	F (%)	(Day s)	F (%)	(Day s)	F (%)	(Day s)	F (%)	(Day s)
C	1.69	58.50	0.98	57.88	1.69	58.50	0.98	57.88	1.69	58.50	0.98	57.88
A1	5.00	19.50	9.95	30.20	2.63	37.50	6.96	34.94	2.56	38.50	4.10	37.02
A2	5.41	18.00	10.86	26.97	4.44	22.00	5.51	32.84	2.22	44.50	2.75	40.88
B	0.00	58.00	2.34	49.99	0.00	58.00	1.74	50.68	0.00	58.00	1.18	51.62
Total	3.35	38.75	6.15	40.10	2.16	47.75	3.62	42.81	1.96	51.25	1.96	46.25

Note. The table presents the adjustment frequency and implied duration for posted, regular, and reference prices. The regular price series results from applying the filter (with $j=7$) presented in Section 3.1 to the posted price series. The reference price series is derived from applying the methodology described in Section 3.2, using a weekly mode, to the posted price series. To aggregate across products, the median and mean of the product values were used. The adjustment frequency is reported as a percentage, while the implied duration is reported in days. It is important to note that the upper limit for the implied duration is constrained by the sample size, which is approximately 60 days. For this reason, implied durations do not exceed 60 days in any case, such as when the adjustment frequency equals zero.

For regular prices, the median adjustment frequencies decrease by approximately 1% to 3% for multichannel supermarkets, while they remain unchanged for the other two sellers, highlighting the limited relevance of these changes for discount stores and price comparison websites. In other words, discount stores could have a greater aggregate impact than what is perceived through posted prices. As for reference prices, the results are very similar to regular prices. Generally, reference prices are only slightly more rigid than regular prices in some cases, a result that contrasts with the findings of Eichenbaum, Jaimovich, & Rebelo (2011), where reference prices are significantly more rigid than regular prices. Since their study uses monthly data and quarterly modes, this highlights the importance of selecting temporary sales when filtering prices, and consequently, its impact on price rigidity.

When considering the adjustment frequency in its mean, it is important to highlight two effects. The first is that due to the maximum implicit duration (around 60 days) because of the data range, the implicit duration might be lower than what it should truly be. The second is

that this aggregation method is more sensitive to extreme values, in this case, very low adjustment frequencies, which result in very high implicit durations. Considering this, the mean adjustment frequencies for posted prices range from about 1% to 11%, showing an implicit duration between 27 and 58 days, or 4 to 9 weeks. For regular prices, the adjustment frequencies decrease by 1% to 7%, with significant differences between multichannel supermarkets and other sellers. In this case, the implicit durations increase by about 1 week compared to posted prices, with durations between 33 and 58 days. Lastly, for reference prices, the prices are approximately 4 days more durable.

To observe variability within and between stores, Figure 1 presents the adjustment frequencies for each product. This distribution allows us to detail aspects that the aggregation measures presented in Table 3 do not capture. For example, while the average adjustment frequency for the A2 supermarket is higher than that for A1, the latter has much higher frequencies for several of its products. One can also easily observe the differences in the distributions, especially in C stores, which make far fewer changes to their product prices compared to other stores. This makes sense with the business model followed by these types of stores, which is based on low prices and limited product variety, allowing them to gain demand within this sector.

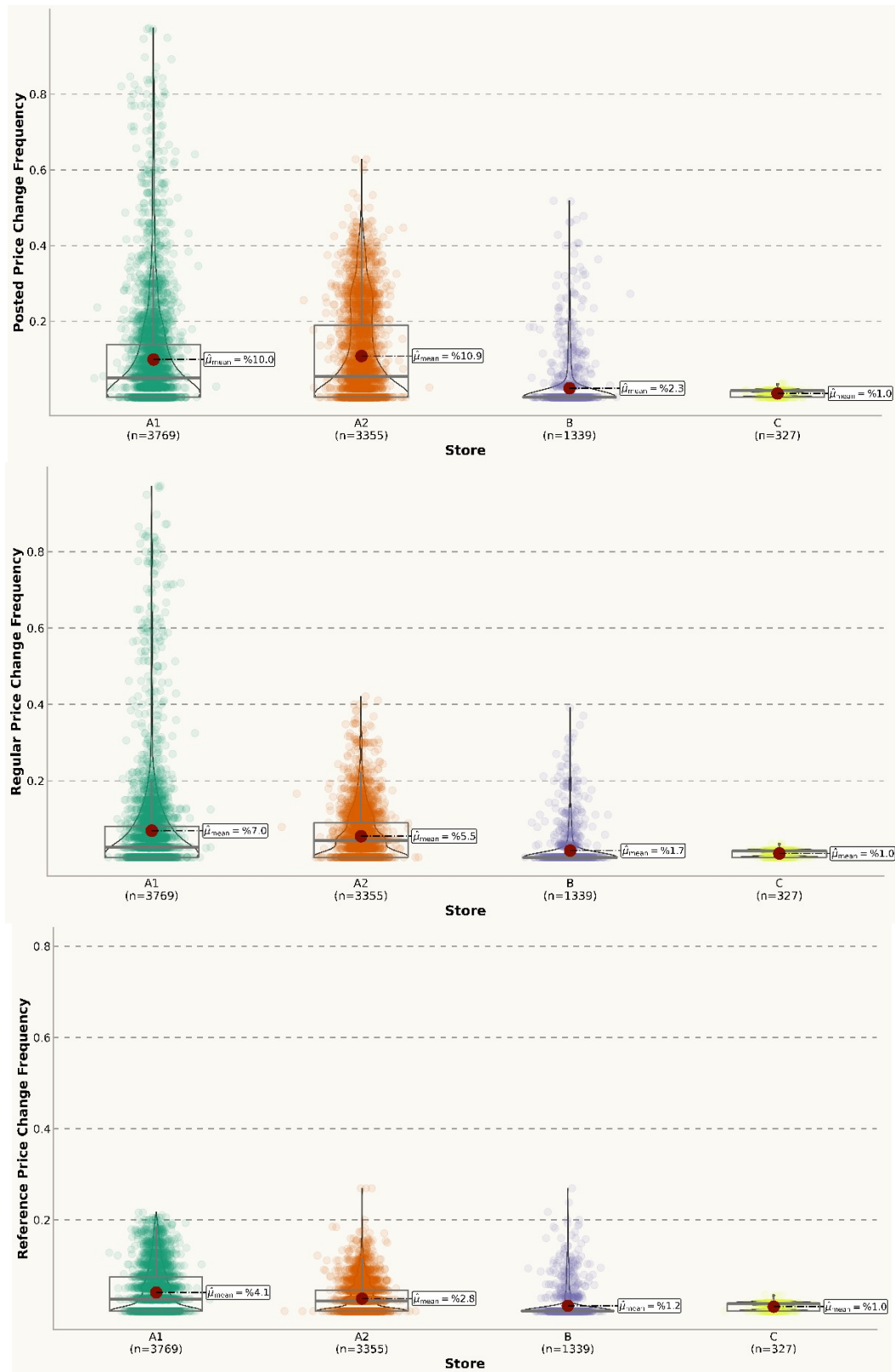


Figure 1A. Adjustment frequency of posted, regular, and reference prices: The figure shows the adjustment frequency of the available products for each store across the three-price series.

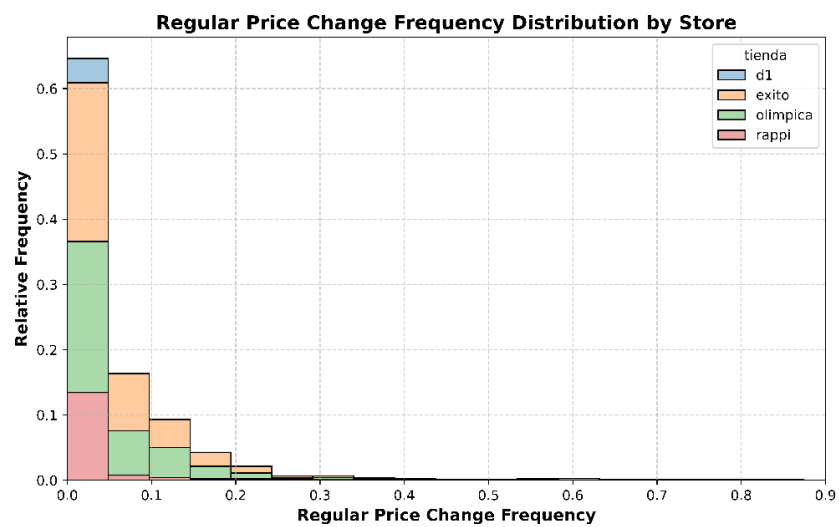
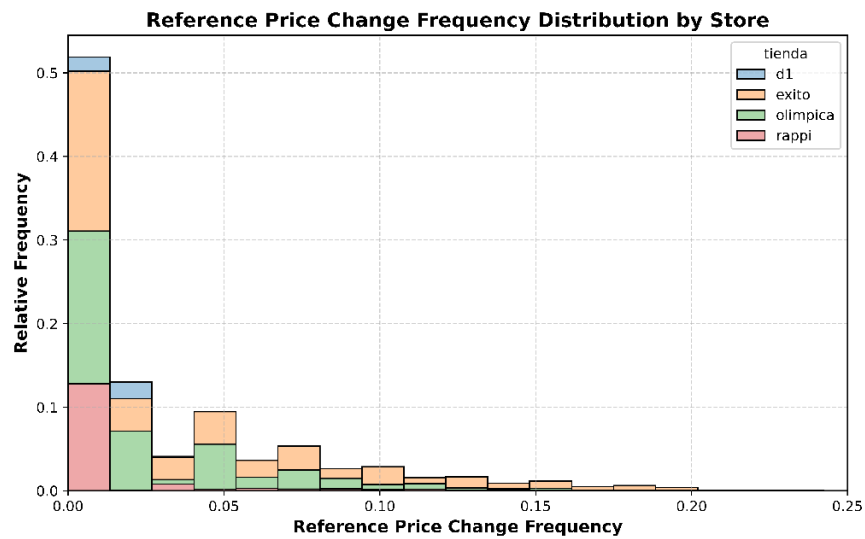
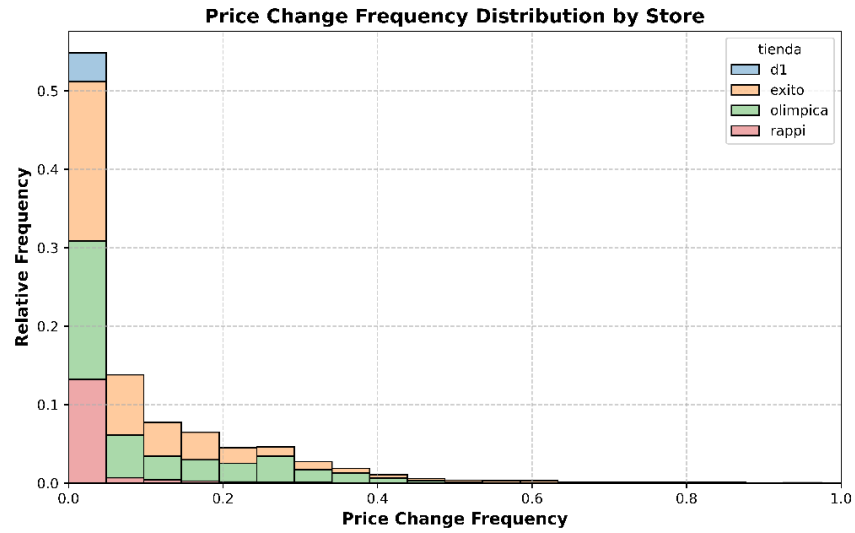


Figure 1B. Frequency of change of posted, regular, and reference prices: The figure shows the adjustment frequency of the available products for each store across the three-price series.

Another important aspect is the difference between A1 and A2, with B having a much lower frequency of price adjustments compared to the other two. This difference can also be explained by the business models, as B is a digital platform that operates as an intermediary, offering products from A1 and A2 in its catalog. This result contrasts with other studies, where a higher frequency of price adjustments is found on price comparison websites compared to other online sellers (Ellison & Ellison, 2009; Gorodnichenko, 2018; Cavallo, 2018). Two possible reasons are identified: The first is that, unlike previous studies, research using data from price comparison websites tends to focus on durable goods, which have higher value and a greater frequency of price adjustments compared to non-durable consumer goods. The second reason relates to differences in the businesses analyzed in these studies, including their business models and market size.

4.1.2. Size and Direction of Adjustment

Regarding the size of the adjustment, there are significant differences between the discount store and the other sellers. This type of store makes price changes with a size smaller than 1% of its price, while for the other businesses, the changes range from 16% to 23%, considering the median adjustment. When considering the mean for the same price series, the adjustments increase, ranging from 1.61% for the discount store to 24.3% for the price comparison store. Just like with the size of the adjustment, the direction is particularly different for the discount store. Price increases are much more common than decreases for the discount store, regardless of the price series or aggregation measure considered. Only when considering the reference prices is the direction of adjustment higher than 50% for all vendors, indicating more price increases than decreases.

Table 4.

Size and direction of adjustment

Seller	Posted				Regular				Reference			
	Median		Mean		Median		Mean		Median		Mean	
	S (%)	Di (%)	S (%)	Di (%)	S (%)	Di (%)	S (%)	Di (%)	S (%)	Di (%)	S (%)	Di (%)
C		100.0		73.8		100.0		73.8		100.0		73.8
	0.28	0	1.61	9	0.28	0	1.61	9	0.28	0	1.61	9
A1	16.2		17.2	48.1	13.8		16.3	49.4	14.8		16.5	52.2
	5	50.00	9	7	8	50.00	1	9	8	50.00	3	0
A2	22.3		22.1	48.2	22.3		20.8	48.3	19.4		20.3	52.2
	1	50.00	0	3	1	50.00	1	7	8	50.00	0	1
B	19.4		24.3	53.9	19.4		24.2	54.1	17.5		19.6	57.4
	4	50.00	0	7	3	50.00	4	1	7	50.00	1	1
Total	17.8		19.7	51.1	16.6		18.5	51.8	16.2		18.0	54.8
	5	50.00	0	0	5	50.00	6	0	2	50.00	7	1

Nota. The size of the adjustment is reported as a percentage; however, it is calculated using a logarithmic approximation (see section 3.4.). The direction of the adjustment is the percentage of price increases over the total changes. For example, the median product for the C store changes by 0.28% in size, and all its changes were positive (100%).

4.1.3. Probability of state change

To enhance the understanding of price adjustments, a transition matrix is proposed for two possible price states: reference or regular, and transitory. This approach aims to determine the probability that a product maintains a specific price type on a given day, conditional on its state the previous day. Table 5 summarizes the transition matrix values between regular and temporary prices for each store, where the entries reflect the probability of transitioning to the state in the column, given that the current state is the one in the row.

For example, in C stores, the probability of shifting to a temporary price (column 2), given that the current state is a reference price (row 1), is 1.8% when using the median and 1.0% when using the mean. Significant differences also arise among the three types of sellers: multichannel large supermarkets have a higher probability of offering temporary prices after

a regular price (approximately 6% and 3% when considering the mean and median, respectively), whereas the discount store and the price comparison website exhibit lower probabilities (between 0% and 2%).

On the other hand, when analyzing the probability of maintaining a temporary price for two consecutive days, the discount store has a probability between 10% and 30% lower than its online counterparts. Likewise, the probability of returning to a reference price after a temporary price is higher for this type of store, suggesting that promotions tend to be shorter-lived. These results align with those found in Section 4.1.1, where price rigidity was observed to be highest in the discount store, followed by the price comparison website, and lowest in multichannel supermarkets.

Table 5.

Transition matrix between reference price and transitory price

Seller	State	Reference Price (1)		Transitory Price (2)	
		Median	Mean	Median	Mean
C	Reference Price (1)	98,2%	99,0%	1,8%	1,0%
A1		97,2%	93,7%	2,8%	6,3%
A2		96,9%	93,3%	3,1%	6,7%
B		100,0%	98,6%	0,0%	1,4%
Total		97,6%	94,5%	2,4%	5,5%
C	Transitory Price (2)	100,0%	90,2%	0,0%	9,8%
A1		66,7%	67,9%	33,3%	32,1%
A2		83,3%	79,7%	16,7%	20,3%
B		83,3%	77,4%	16,7%	22,6%
Total		75,0%	74,5%	25,0%	25,5%

Note: To aggregate the values at the store level, the transition matrix was calculated at the product level, and then the mean and median of the values for each store were computed.

If the series of regular prices is considered instead of reference prices, the results remain similar (see Table 6). The stores with the highest probability of price changes are the

multichannel large supermarkets, followed by the price comparison website, and lastly, the discount store. Most likely, due to the time window used in this analysis, no price changes were recorded for the discount store. This point will be revisited later, where a robustness check will be conducted using a larger sample with a longer time window.

Table 6.

Transition matrix between regular price and transitory price

Seller	State	Regular Price (1)		Transitory Price (2)	
		Mediana	Media	Mediana	Media
C	Regular price (1)	100,0%	100,0%	0,0%	0,0%
A1		100,0%	98,1%	0,0%	1,9%
A2		100,0%	96,9%	0,0%	3,1%
B		100,0%	99,6%	0,0%	0,4%
Total		100,0%	97,9%	0,0%	2,1%
C	Transitory Price (2)	-	-	-	-
A1		50,0%	62,6%	50,0%	37,4%
A2		100,0%	85,7%	0,0%	14,3%
B		100,0%	76,5%	0,0%	23,5%
Total		100,0%	75,4%	0,0%	24,6%

Note: To aggregate the values at the store level, the transition matrix was calculated at the product level, and then the mean and median of the values for each store were computed.

Another important aspect for understanding price dynamics in online markets is the conditional probabilities based on price values in different past periods. These probabilities help analyze price adjustment dynamics by providing information on the current price state relative to past prices. For example, an interesting measure is the conditional probability of switching between states over a week or a month. Additionally, this approach helps determine whether prices stabilize over time and, if so, at what values. These convergence values provide insights into the long-term behavior of price changes.

Figure 2 shows the changes in transition probabilities between regular and reference states, broken down by each store. Two interesting dynamics can be distinguished: the decrease in the probability of maintaining the same price state, whether regular or transitory, and the increase in the probability of switching between price states. For example, the probability that the price today is regular, given that it was regular 7 days ago, is higher than the probability that this occurs when the regular price was set 30 days ago. Also, the probability that the price changes its state, for example, from reference to transitory, is lower if the initial state was observed 7 days ago compared to 30 days ago.

Just like with the adjustment frequencies between stores, the values for the transition probabilities show differences between vendors, which vary depending on the aggregation method. The C supermarket and B show the highest probabilities of changing to a reference price state, which aligns with the results in the adjustment frequencies. However, when analyzing the dynamics, the price comparison website shows a distinctive characteristic: it shows a jump between the first and second periods when the initial state is a reference price. This indicates a significant change in the probability after the first day when the initial state is a reference price, which can be associated with a change in the persistence of this type of price after this period. In other words, if the price of a product today is the reference price, it is likely to remain the same the next day; however, by the second day, this probability drops considerably.

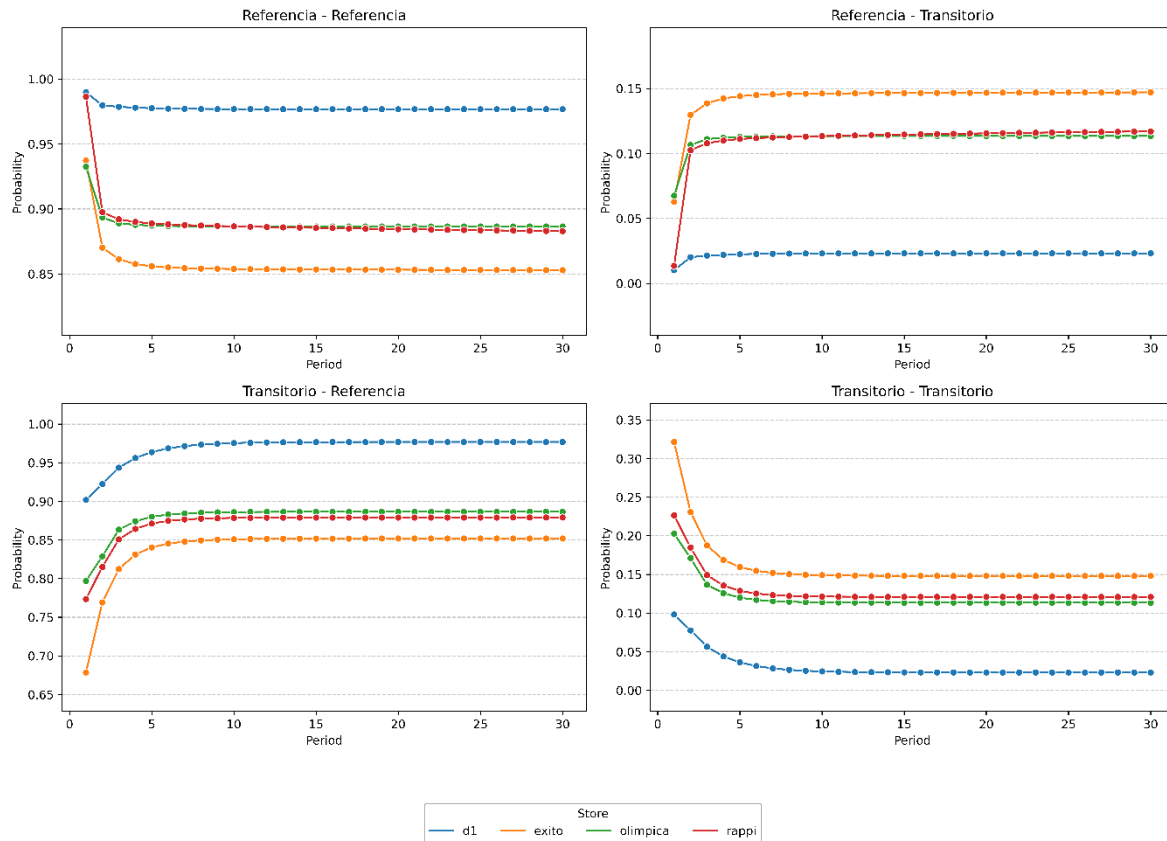


Figure 2. Transition probabilities between states by store: The probability of transition between the two states is presented as a function of time periods. The convergence of probabilities is reached in less than 30 days, indicating independence between price states after this number of days.

Figure 3 presents the results of the transition probabilities between states when considering the regular price series. Since this filter removes most transitory prices in store C, the probability of being in or transitioning to that state is always zero. This series highlights the distinct behavior of store A1, which shows a greater predominance of transitory prices. As a result, it differs from other sellers in two key aspects: first, a lower probability of maintaining regular prices, which stabilizes at around 87% after the second week. Second, it exhibits the highest probability of transitioning from a regular price to a transitory one in the long run, stabilizing at approximately 11%. These findings offer a deeper understanding of price adjustments in online markets—since the probabilities stabilize within approximately two

weeks, they complement the single-period transition matrix by incorporating long-term dynamics.

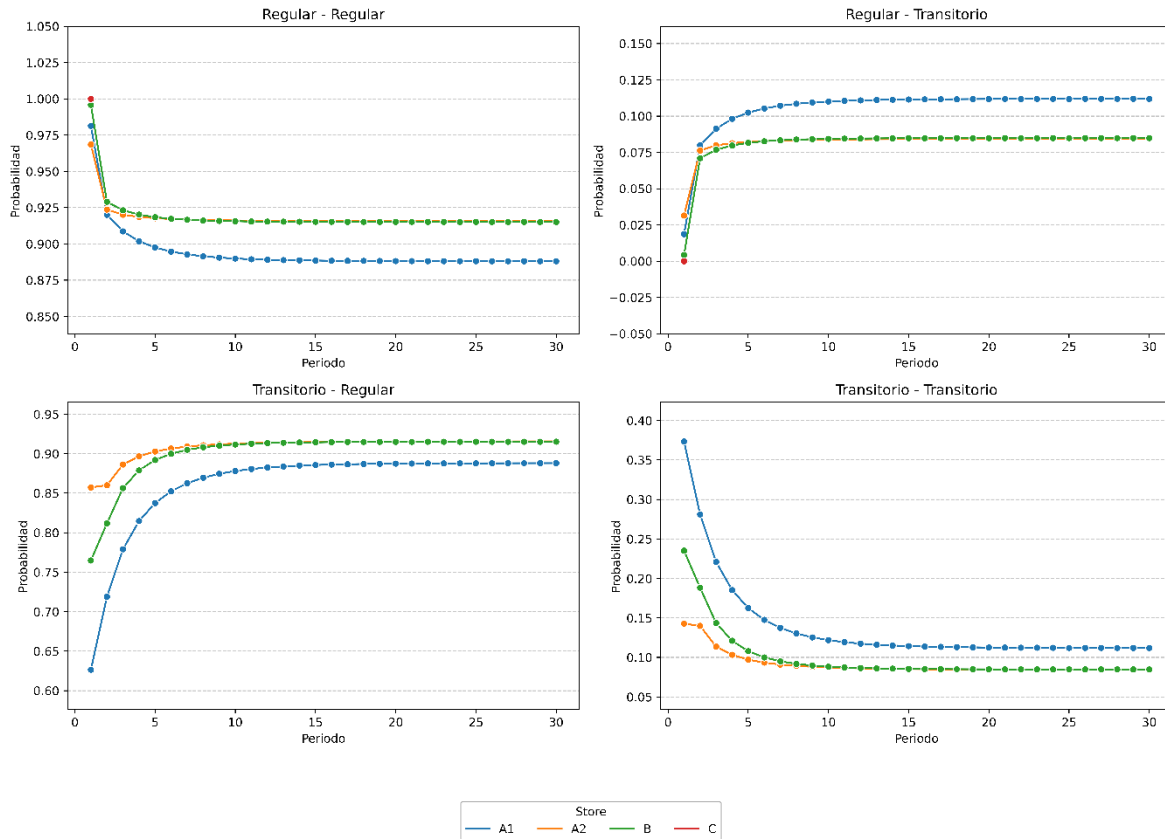


Figure 3. Transition Probabilities Between Regular and Transitory Prices by Store: This figure presents the probability of transitioning between the two price states over different time periods. The convergence of probabilities is achieved in less than 30 days, indicating that price states become independent beyond this time frame.

4.1.4. Product categories

Due to the existing differences in the products available according to vendors and the possible variations in price adjustments depending on the characteristics of these products, a classification of the products was made according to the divisions of the Classification of Individual Consumption by Purpose - COICOP. This classification helps control for the differences between products, allowing us to determine whether the differences in price

adjustment across vendors remain consistent across purchase categories. Due to the nature of the data, where it is not possible to access a unique identification code or any other element that would allow for automatic classification, the decision was made to manually classify the unique products and create a dictionary with these classifications. This presents a greater challenge than in other types of data, such as those used in the CPI.

By extracting the product categories, significant differences between vendors were found. For example, for the first category: Alcoholic Beverages, Tobacco, and Narcotics, the vendor A2 shows a price adjustment frequency of 16% and an implicit duration of only 5.63 days, when considering the median as the aggregation method and the posted price series. This behavior is quite different from the other vendors, whose implicit durations range from 49 to 58 days. For the same store, category, and aggregation method, the price adjustment frequencies decrease by approximately 14% when considering reference prices, reflecting an implicit duration of 41 days. As seen, the implicit duration increases from less than one week to almost 7 weeks when reference prices are considered, indicating a significant inertia in the prices.

In general, the differences between vendors persist when considering the product categories, price series, and aggregation methods. For most categories, multichannel supermarkets show a higher frequency of price adjustments compared to price comparison websites and discount stores, results that align with those found in section 4.1.1. This supports the hypothesis about the significant differences in price adjustments in online markets, especially between multichannel supermarkets and the hard discount store, which has a very low-price adjustment frequency. This could potentially allow for a greater real effect of monetary policies.

Table 6.
Frequency and Duration of Prices by Purchase Categories

Category	Seller	Products	Posted				Regular				Reference			
			Median		Mean		Median		Mean		Median		Mean	
			F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)
Alcoholic Beverages, Tobacco And Narcotics	C	31	1.75	56.50	1.63	57.08	1.75	56.50	1.63	57.08	1.75	56.50	1.63	57.08
	A1	978	0.00	49.00	4.44	38.92	0.00	59.00	3.29	41.83	0.00	59.00	2.90	41.96
	A2	608	16.28	5.63	15.59	19.14	4.76	20.50	6.80	26.26	2.38	41.50	3.81	35.04
	B	612	0.00	58.00	1.70	52.21	0.00	58.00	1.15	52.89	0.00	58.00	0.68	53.62
	Total	2221	0.00	53.00	6.69	37.43	0.00	56.00	3.64	40.83	0.00	56.00	2.52	43.48
Clothing And Footwear	C	-	-	-	-	-	-	-	-	-	-	-	-	-
	A1	53	0.00	53.00	5.39	42.03	0.00	53.00	4.47	43.27	0.00	55.00	2.16	44.86
	A2	13	28.00	3.04	28.08	13.10	23.08	3.81	20.08	13.79	10.53	8.99	12.37	20.11
	B	-	-	-	-	-	-	-	-	-	-	-	-	-
	Total	66	0.00	47.00	9.86	36.33	0.00	47.00	7.54	37.46	0.00	53.00	4.17	39.98
Food And Non-Alcoholic Beverages	C	290	1.69	58.50	0.91	58.05	1.69	58.50	0.91	58.05	1.69	58.50	0.91	58.05
	A1	1383	10.00	9.49	12.24	20.59	5.00	19.50	6.67	29.22	5.00	19.50	4.81	32.03
	A2	2004	5.00	19.50	10.64	27.75	2.63	37.50	5.46	33.74	0.00	56.00	2.35	43.05
	B	666	0.00	58.00	2.85	48.65	0.00	58.00	2.20	49.31	0.00	58.00	1.52	50.50
	Total	4343	4.76	20.50	9.30	30.70	2.50	39.50	5.04	36.31	0.00	55.00	2.91	41.68
Furnishings, Household Equipment And Routine Household	C	-	-	-	-	-	-	-	-	-	-	-	-	-
	A1	647	4.55	21.50	15.27	28.33	2.86	34.50	13.69	30.66	2.70	36.50	5.28	34.28
	A2	356	2.33	42.50	5.33	34.25	2.33	42.50	4.17	36.11	2.27	43.50	2.72	40.78
	B	2	3.85	36.75	3.85	36.75	3.85	36.75	3.85	36.75	3.85	36.75	3.85	36.75
	Total	999	2.94	33.50	11.73	30.44	2.70	36.50	10.30	32.60	2.56	38.50	4.37	36.59

Categories	Seller	Products	Posted				Regular				Reference			
			Median		Mean		Median		Median		Mean		Median	
			F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)
Information And Communication	C	-	-	-	-	-	-	-	-	-	-	-	-	-
	A1	68	15.89	5.84	29.83	20.25	15.16	6.24	29.25	20.29	9.05	10.66	9.46	23.52
	A2	96	1.11	50.75	4.64	36.41	0.00	58.00	3.38	40.73	0.00	57.00	2.84	40.00
	B	3	0.00	55.00	4.00	40.27	0.00	55.00	1.33	45.83	0.00	55.00	1.33	45.83
	Total	161	5.00	19.50	14.88	29.90	2.70	36.50	13.88	32.50	2.86	34.50	5.51	33.42
Personal Care, Social Protection And Miscellaneous Goods And Services	C	1	0.00	58.00	0.00	58.00	0.00	58.00	0.00	58.00	0.00	58.00	0.00	58.00
	A1	182	2.74	36.00	11.92	31.01	2.60	38.00	8.24	33.14	0.00	55.00	5.03	36.64
	A2	116	9.09	10.49	15.05	15.97	6.82	14.16	7.91	24.21	4.44	22.00	4.15	32.54
	B	31	0.00	52.00	4.53	38.73	0.00	52.00	3.39	40.04	0.00	53.00	3.13	40.73
	Total	330	6.82	14.16	12.29	26.53	4.44	22.00	7.64	30.72	2.50	39.50	4.53	35.63
Recreation, Sport And Culture	C	4	1.72	57.50	1.72	50.87	1.72	57.50	1.72	50.87	1.72	57.50	1.72	50.87
	A1	419	0.00	55.00	3.96	42.63	0.00	59.00	2.55	45.25	0.00	59.00	1.86	46.40
	A2	133	0.00	57.00	5.34	38.93	0.00	57.00	2.55	42.77	0.00	57.00	2.11	43.53
	B	4	0.00	58.50	0.00	56.75	0.00	58.50	0.00	56.75	0.00	58.50	0.00	56.75
	Total	551	0.00	57.00	4.24	41.91	0.00	58.00	2.52	44.79	0.00	58.00	1.91	45.83
Transport	C	-	-	-	-	-	-	-	-	-	-	-	-	-
	A1	20	0.00	59.00	0.00	58.95	0.00	59.00	0.00	58.95	0.00	59.00	0.00	58.95
	A2	2	5.93	16.91	5.93	16.91	5.93	16.91	5.93	16.91	3.60	31.25	3.60	31.25
	B	-	-	-	-	-	-	-	-	-	-	-	-	-
	Total	22	0.00	59.00	0.54	55.13	0.00	59.00	0.54	55.13	0.00	59.00	0.33	56.43

Note. As in section 4.1.1, the implicit durations are capped from above based on the availability of days for each product. In any case when the adjustment frequency is zero, the implicit duration will be equal to the number of days of information available for the given product.

4.2. Robustness Analysis

As a robustness check on the results, the statistics calculated in Section 4.1 are re-estimated using a dataset with a larger number of days and products, but without restricting the comparison to the same time period—that is, sellers may have different data collection periods. For comparability, the same filters used in the previous section are applied to this dataset.

4.2.1. Frequency of adjustment and Implied Duration

The results suggest a longer implicit price duration for the discount store when using a broader time window. As previously mentioned, the results in the previous section showed a limit close to 60 days, meaning that durations did not exceed this value in any case. By including a larger number of days in the sample, the discount store significantly increased the implicit duration of its prices, rising from approximately 58 to 102 days. Large multichannel supermarkets remained within the range of the previous results, reflecting a much higher price adjustment frequency compared to both the discount store and the price comparison website.

Price adjustment frequencies in large multichannel supermarkets are 5 to 20 times higher than in the discount store for posted prices and 3 to 4 times higher when compared to the price comparison website (using the mean as an aggregation method). The results align with those presented in the previous section: in general, the discount store exhibits the most rigid prices among sellers when considering the mean. However, when using the median, the price comparison website shows the highest price rigidity.

It is important to emphasize that it cannot be concluded that prices are more rigid on the price comparison website than in the discount store, despite the median aggregation suggesting this. This is because the data collection window for the discount store is shorter, making it impossible to determine the exact duration. It can only be concluded that the duration exceeds its temporal window (60 days). Unlike findings in previous literature, such as Cavallo (2018) and Ellison & Ellison (2018), this study found a higher price adjustment frequency in large multichannel supermarkets than on the price comparison website.

Cavallo (2018) estimated an implicit price duration of 2.29 months for Colombia's online market based on posted prices and 2.6 months after removing promotional prices, using data from a single retailer between 2007 and 2010. In contrast, this study finds lower implicit price durations between 2022 and 2023, ranging from 1.48 to 1.6 months for posted prices, 1.66 to 1.7 months for regular prices, and 1.7 to 1.76 months for reference prices, depending on the aggregation method used. It is important to highlight the considerable differences across business models. Price duration varies between 0.38 and 3.4 months depending on the online store analyzed, ranging from 11 to 103 days for A1 and C, respectively. These estimates suggest lower online price rigidity in the Colombian market and highlight the significant differences between various online sellers.

Table 8.
Frequency of adjustment and Implied Duration Full Sample

Seller	Posted				Regular				Reference			
	Median		Mean		Median		Mean		Median		Mean	
	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)
C	0.97	101.50	0.59	103.72	0.97	101.50	0.59	103.72	0.97	101.50	0.59	103.72
A1	4.55	21.50	9.25	38.26	3.03	32.50	6.53	44.10	2.86	34.50	4.17	45.96
A2	8.22	11.66	12.12	28.89	5.41	18.00	7.30	34.79	3.28	30.00	4.35	43.59
B	0.00	67.00	3.48	57.52	0.00	67.00	2.71	58.25	0.00	67.00	1.76	59.49
Total	2.76	44.25	6.37	47.89	2.00	49.75	4.62	51.17	1.91	50.75	2.96	52.73

Note. The table presents the frequency of adjustment and implied duration for posted, regular, and reference prices. The regular price series result from applying the filter (with $j = 7$) presented in section 3.1 to the posted price series. The reference price series are the result of applying the methodology from section 3.2 with a weekly mode to the posted price series. To aggregate across products, the median and the mean values for the products were used. The frequency of adjustment is reported as a percentage, while the implied duration is reported in days. When the frequency of adjustment was zero, the implied duration reflects the total number of days available in the sample.

The results for the differences between price series remain consistent when using the full dataset. In general, reference prices are slightly more rigid than regular prices, and both are more rigid than posted prices, a finding consistent with the literature. In this case, regular prices exhibit a duration approximately one week longer than posted prices. Similarly, reference prices have a duration about one week longer than regular prices.

When analyzing the full distribution in Figure 4, it is evident that large multichannel supermarkets have similar mean and median values, with broader distribution ranges compared to the discount store and the price comparison website. For the latter, the significance of outliers stands out, which is reflected in an atypical mean value that lies outside the box in the box plot.

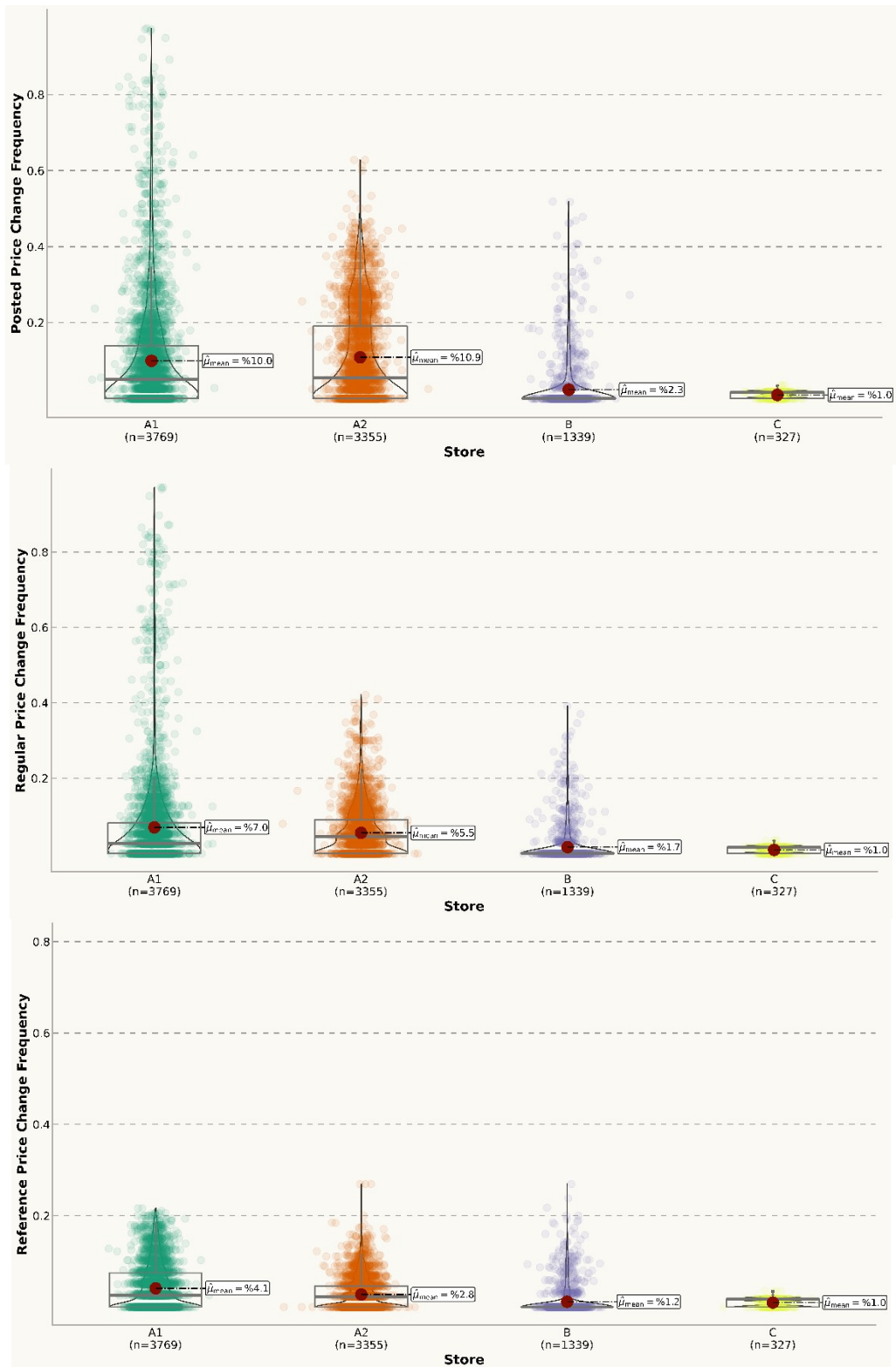


Figure 4A. Adjustment frequency of posted, regular, and reference prices: The figure shows the adjustment frequency of the available products for each store across the three-price series.



Figure 4B. Frequency of adjustment for posted, regular, and reference prices: The figure shows the frequency of adjustment for the products available in each store across the three price series.

4.2.2. Size and Direction of Adjustment

The results for the size and direction of price adjustments are very similar to those in the main analysis. The discount store exhibits a significantly smaller adjustment size compared to its online counterparts. The direction of adjustment is close to 50% for most sellers, suggesting a similar number of price increases and decreases. However, the discount store deviates from this pattern across all price series, with 71.53% of adjustments being price increases.

Table 9.

Size and direction of adjustment Full Sample

Seller	Posted				Regular				Reference			
	Median		Mean		Median		Mean		Median		Mean	
	S (%)	Di (%)	S (%)	Di (%)	S (%)	Di (%)	S (%)	Di (%)	S (%)	Di (%)	S (%)	Di (%)
C	0.31	100.00	2.18	71.53	0.31	100.00	2.18	71.53	0.31	100.00	2.18	71.53
A1	16.03	50.00	17.73	46.81	13.70	50.00	16.42	46.30	14.56	50.00	16.90	46.59
A2	21.22	50.00	21.23	47.08	19.31	50.00	19.90	46.87	16.43	50.00	18.81	47.15
B	19.36	50.00	23.24	53.99	18.37	50.00	23.19	54.27	17.60	50.00	20.97	55.24
Total	17.69	50.00	19.48	50.53	16.04	50.00	18.16	50.57	15.49	50.00	17.86	51.20

Nota. The adjustment size is reported as a percentage; however, it is calculated using a logarithmic approximation (see Section 3.4). The direction of adjustment represents the percentage of price increases over the total number of changes. For example, the median product for store C has an adjustment size of 0.31%, with 71.53% of its price changes being increases.

4.2.3. Probability of state change

The probabilities of price adjustments remain largely unchanged when considering the full dataset. The results suggest that the sellers with the highest probability of maintaining a reference price are the discount store and the price comparison website, with probabilities ranging between 98% and 100%, depending on the aggregation method. Large multichannel

supermarkets are the most likely to offer and maintain transitory prices. The probability of maintaining a transitory price for these sellers ranges between 18% and 33%. Additionally, the probability of switching from a reference price to a transitory price is between 2.5% and 6.8% for these sellers, compared to a range of 0% to 1.9% for the others.

Table 10.
Transition matrix between reference price and transitory price Full Sample

Seller	State	Reference Price (1)		Transitory Price (2)	
		Median	Mean	Median	Mean
C	Reference Price (1)	99,3%	99,4%	0,7%	0,6%
A1		97,5%	94,4%	2,5%	5,6%
A2		95,7%	93,2%	4,3%	6,8%
B		100,0%	98,1%	0,0%	1,9%
Total		97,9%	94,6%	2,1%	5,4%
C	Transitory Price (2)	100,0%	85,4%	0,0%	14,6%
A1		66,7%	67,9%	33,3%	32,1%
A2		81,8%	79,6%	18,2%	20,4%
B		100,0%	80,8%	0,0%	19,2%
Total		71,4%	73,5%	28,6%	26,5%

Note: To aggregate the values at the store level, the transition matrix was calculated at the product level, and then the mean and median of the values for each store were computed.

When considering the regular price series, the probabilities of transitioning to a transitory price are nearly zero for all sellers. Analyzing this price series, it is observed that the discount store never had a transitory price, reflecting the low prevalence of promotions in this type of retailer. Similar to reference prices, large multichannel supermarkets are the most likely to offer transitory prices after having regular prices, although with a relatively low probability—ranging between 1.7% and 2.8% when using the mean as the aggregation method, and 0% when using the median.

Table 11.

Transition matrix between regular price and transitory price Full Sample

Seller	State	Regular Price (1)		Transitory Price (2)	
		Mediana	Media	Mediana	Media
C	Regular price (1)	100,0%	100,0%	0,0%	0,0%
A1		100,0%	98,3%	0,0%	1,7%
A2		100,0%	97,2%	0,0%	2,8%
B		100,0%	99,5%	0,0%	0,5%
Total		100,0%	98,2%	0,0%	1,8%
C	Transitory Price (2)	-	-	-	-
A1		50,0%	61,8%	50,0%	38,2%
A2		100,0%	83,9%	0,0%	16,1%
B		100,0%	82,0%	0,0%	18,0%
Total		80,0%	72,3%	20,0%	27,7%

Note: To aggregate the values at the store level, the transition matrix was calculated at the product level, and then the mean and median of the values for each store were computed.

Regarding the probabilities conditioned on longer periods, a considerable change can be observed for the price comparison website (see Figure 5). It is not surprising that this result is similar to the one in the previous section, as the sample for this seller exhibited the least variation, increasing from 61 to 83 days and from 1,339 to 1,914 products (see Table 1). However, the results for other sellers also remain relatively constant, with a rapid convergence, generally within the first week.

Once again, the discount store is the one whose probabilities change the least as the number of periods increases. Large online supermarkets converge to similar values among themselves, along with the price comparison website. Despite starting at different probability values, large online supermarkets and the price comparison website converge to very similar values after two weeks, suggesting a fairly similar behavior in the long run.

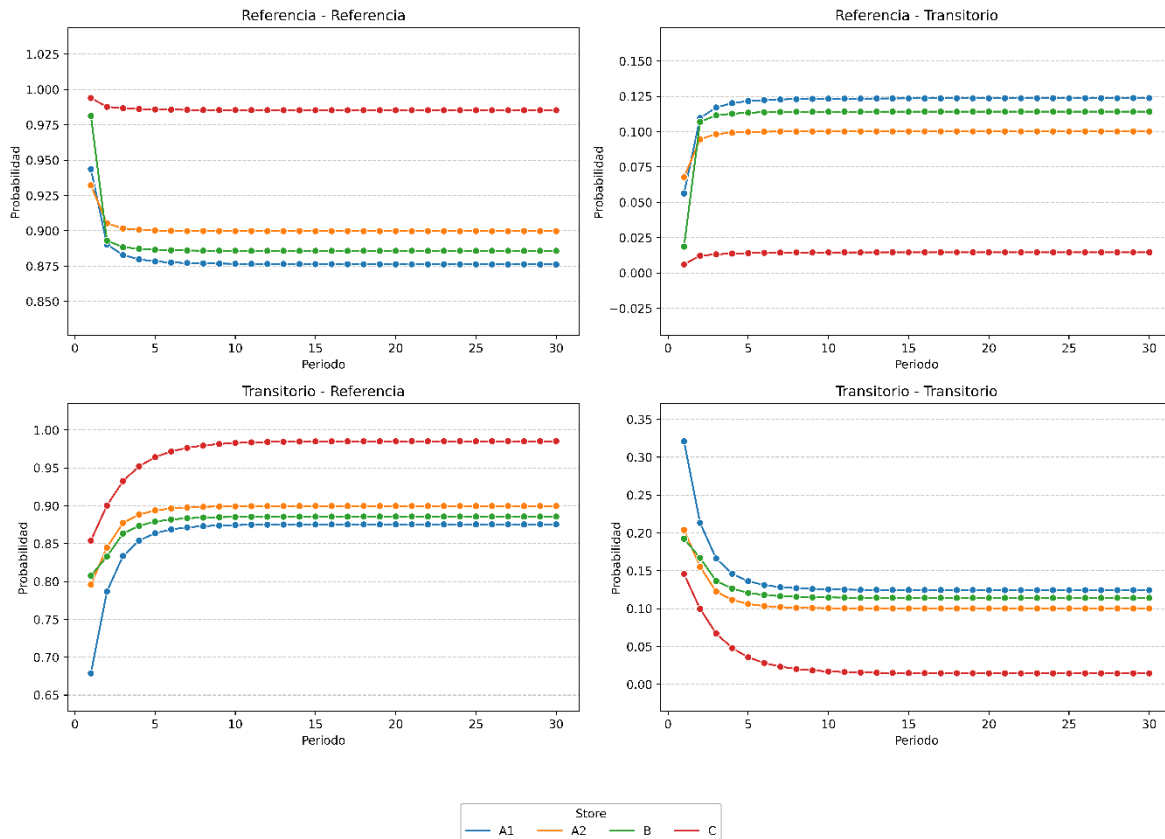


Figure 5. Transition probabilities between reference and transitory prices by store for the entire sample: The transition probability between the two states is presented as a function of time periods. The convergence of the probabilities is reached in less than 30 days, indicating independence between price states after this number of days.

When using the full dataset and considering the regular price series, the probability values once again converge between the sellers, except for the discount store, which did not have prices in a transitory state. Similarly to the reference prices, the probability of transitioning between regular and transitory prices converges after the second week. It can be observed that the behavior on the first day for one of the sellers is particularly distinct, especially when their initial state is transitory (bottom row). However, after approximately 10 periods, their behavior closely aligns with that of the other sellers.

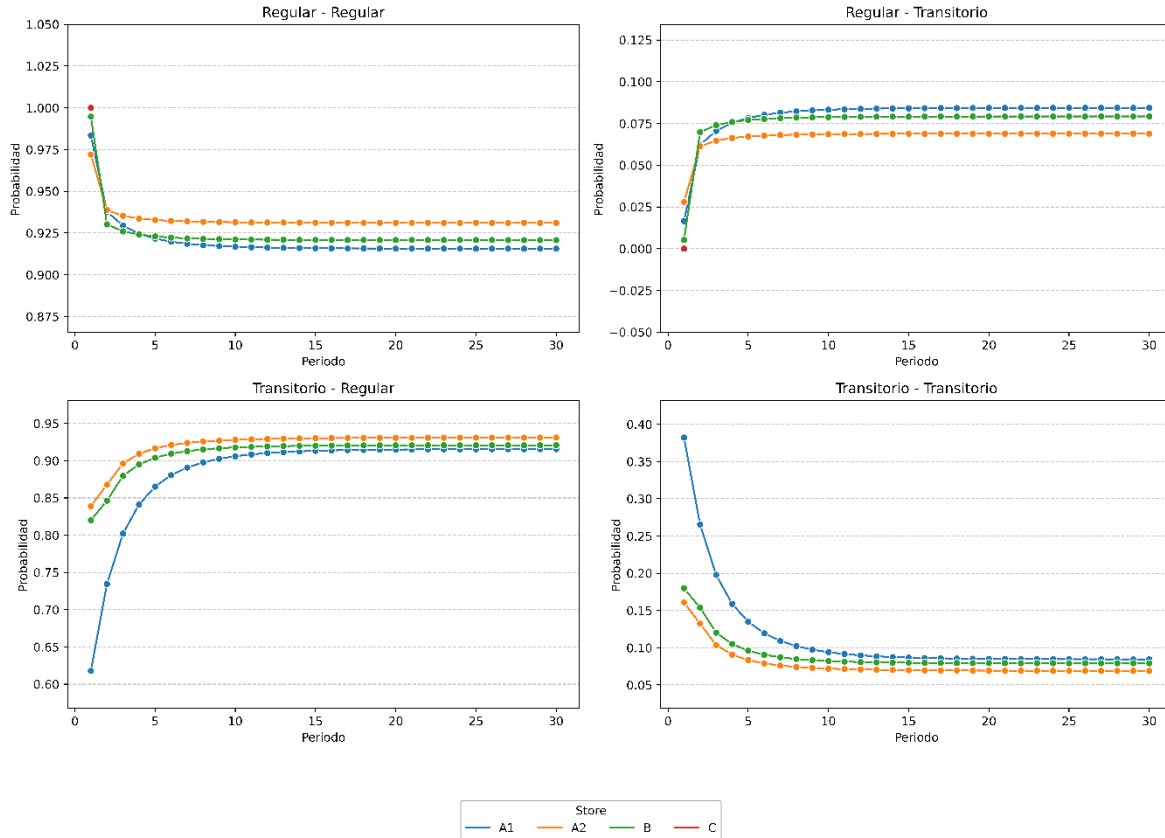


Figure 6. Transition probabilities between regular and transitory prices by store: The transition probability between the two states is presented as a function of time periods. The convergence of the probabilities is reached in less than 30 days, indicating independence between price states after this number of days.

4.2.4. Product categories

Using the full product sample, the results from the previous section are expected to be complemented, allowing for higher values over 60 days for the implicit durations. By using this sample, it is found that the categories with the longest implicit price durations, when considering the median aggregation method, are Transportation and Leisure, Sports, and Culture, with durations of 59 and 91 days, respectively, when considering the posted price series. On the other hand, the categories with the shortest duration using the same aggregation method and price series are Personal Care and Health, with durations of 12.69 and 17 days, respectively.

When contrasting these results with those obtained using the subsample of goods, the results are consistent. The values for each store remain consistent across categories, reflecting certain differences based on product types. For example, for one of the categories with the highest number of goods, Non-alcoholic Beverages and Food, it is confirmed that the price duration is less than two weeks in large multichannel supermarkets, while in the discount store, it reaches 102 days. Differences in price adjustments according to the type of seller persist, however, these results should be taken with caution, especially due to the number of products in some of these categories. The categories with fewer products were Transportation, Health, Clothing, and Footwear, with between 22 and 66 products.

Frequency and Duration of Prices by Purchase Categories Full Sample

Category	Seller	Products	Posted				Regular				Reference			
			Median		Mean		Median		Mean		Median		Mean	
			F (%)	(Days)	F (%)	d	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	d
Alcoholic Beverages, Tobacco And Narcotics	C	31	0.98	101.50	0.89	120.2	0.98	101.5	0.89	120.21	0.98	101.5	0.89	120.2
	A1	978	3.08	32.00	4.67	49.28	1.55	61.00	3.10	58.21	1.64	60.50	2.93	58.21
	A2	609	14.6	6.30	14.7	17.47	6.67	14.49	7.74	24.49	4.41	22.16	5.00	24.49
	B	612	0.00	68.00	1.83	63.24	0.00	68.00	1.28	63.98	0.00	68.00	0.78	63.98
	Total	2221	2.88	34.25	6.60	45.41	1.56	56.50	3.84	51.45	1.52	59.75	2.88	51.45
Clothing And Footwear	C	-	-	-	-	-	-	-	-	-	-	-	-	-
	A1	53	0.00	68.00	4.80	55.81	0.00	68.00	3.92	56.82	0.00	68.00	2.19	56.82
	A2	14	21.2	4.19	25.3	14.60	20.7	4.29	18.5	15.52	11.68	8.07	12.3	15.52
	B	-	1	-	4	-	9	-	6	-	-	-	2	-
	Total	66	1.89	52.50	9.09	47.20	1.89	52.50	6.98	48.19	0.00	54.00	4.31	48.19
Food And Non- Alcoholic Beverages	C	290	0.62	102.50	0.57	106.0	0.62	102.5	0.57	106.02	0.62	102.5	0.57	106.02
	A1	1384	9.09	10.49	11.81	25.03	4.69	20.83	6.52	36.71	4.08	24.00	4.81	36.71
	A2	2004	8.00	11.99	12.1	28.66	5.56	17.50	7.34	33.62	3.23	30.50	3.97	33.62
	B	666	0.00	67.00	2.95	58.06	0.00	67.00	2.25	58.77	0.00	67.00	1.58	58.77
	Total	4343	5.59	17.40	9.86	37.18	3.39	29.00	5.85	43.29	2.70	36.50	3.64	43.29
Furnishings , Household Equipment And Routine Household	C	-	-	-	-	-	-	-	-	-	-	-	-	-
	A1	649	4.76	20.50	15.11	34.95	3.33	29.50	13.3	38.85	2.56	38.50	5.00	38.85
	A2	357	3.57	27.50	5.86	44.68	3.08	32.00	4.74	46.90	1.89	52.50	3.13	46.90
	B	2	3.33	41.25	3.33	41.25	3.33	41.25	3.33	41.25	3.33	41.25	3.33	41.25
	Total	999	3.77	26.00	11.81	38.41	3.31	29.75	10.3	41.70	2.44	40.50	4.33	41.70
Health	C	1	0.00	88.00	0.00	88.00	0.00	88.00	0.00	88.00	0.00	88.00	0.00	88.00
	A1	19	2.38	41.50	13.4	47.94	2.38	41.50	6.08	48.88	2.38	41.50	4.79	48.88
	A2	27	23.1	3.79	20.1	12.28	5.36	18.16	4.49	34.58	2.82	35.00	2.60	34.58
	B	21	0.00	65.00	1.72	51.44	0.00	65.00	1.72	51.44	0.00	65.00	1.72	51.44
	Total	48	3.15	39.75	5.83	49.91	2.77	40.38	5.33	48.96	2.22	41.50	3.11	48.96

Categories	Seller	Products	Posted		Regular				Referen			
			Median		Mean		Median		Median		Mean	
			F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)
Information And Communication	C	-	-	-	-	-	-	-	-	-	-	-
	A1	69	12.07	7.77	28.88	23.30	12.07	7.77	28.42	23.33	7.14	13.49
	A2	97	1.75	56.50	4.41	54.75	1.47	67.50	3.59	58.95	1.52	65.50
	B	3	0.00	55.00	4.94	43.08	0.00	55.00	2.47	45.33	0.00	55.00
	Total	161	5.00	19.50	14.41	41.70	3.12	31.50	13.71	44.17	3.12	31.50
Personal Care, Social Protection And Miscellaneous Goods And Services	C	1	0.00	109.00	0.00	109.00	0.00	109.00	0.00	109.00	0.00	109.00
	A1	193	3.64	27.00	10.46	41.17	3.23	30.50	6.84	44.00	2.33	42.50
	A2	116	10.91	8.66	16.35	17.12	8.15	11.76	9.31	22.48	4.58	21.33
	B	31	0.00	55.00	4.70	42.59	0.00	55.00	3.60	43.94	0.00	55.00
	Total	330	7.58	12.69	11.91	33.31	4.92	19.83	7.36	36.86	3.23	30.50
Recreation, Sport And Culture	C	4	0.97	102.50	1.04	90.62	0.97	102.50	1.04	90.62	0.97	102.50
	A1	420	0.94	59.00	4.35	62.48	0.00	59.00	3.22	64.54	0.00	59.00
	A2	133	2.74	36.00	5.61	50.82	1.82	54.50	3.47	54.39	1.82	54.50
	B	4	0.00	66.00	0.00	62.75	0.00	66.00	0.00	62.75	0.00	66.00
	Total	551	1.47	59.00	4.59	59.92	1.25	59.00	3.24	62.31	1.16	59.00
Transport	C	-	-	-	-	-	-	-	-	-	-	-
	A1	20	0.00	91.00	0.11	85.22	0.00	91.00	0.11	85.22	0.00	91.00
	A2	2	4.55	22.83	4.55	22.83	4.55	22.83	4.55	22.83	2.67	40.50
	B	-	-	-	-	-	-	-	-	-	-	-
	Total	22	0.00	91.00	0.52	79.55	0.00	91.00	0.52	79.55	0.00	91.00

Note. The number of products does not vary as much as reported in Table 1, since the products were manually classified to perform the main analysis. However, the number of observations increased due to the larger time window used, which allowed for a broader data coverage. This may have contributed to the capture of price dynamics over time and greater accuracy in calculating implicit durations and transition probabilities.

5. Conclusions

Online markets are becoming increasingly important within the global economy, transforming the conditions faced by agents. They allow consumers to monitor the prices of various products almost instantaneously, reducing monitoring and price search costs. On the other hand, they also change the way producers set prices, as they open up the possibility of automating price changes and even monitoring competition more easily. However, it is important to consider the seller's business model.

It is found that the speed and size of price adjustments vary significantly within the same online market for the same region. Specifically, the results suggest that the seller's business model is a key feature in understanding the speed and size of price adjustments within online markets. Implicit durations can be between 5 and 20 times longer for some types of businesses, suggesting that online prices can be more rigid for certain sellers who seek to offer low prices for a limited number of products. Nonetheless, prices in physical markets are more rigid than prices in virtual stores, suggesting that menu costs are one of the most important sources of rigidity in physical markets.

The data collected through web scraping offers the possibility of complementing traditional sources for measuring price rigidity, providing interesting characteristics such as easy collection, high frequency, and real-time access to direct information from sellers. With this, the ability to estimate price rigidity in virtual markets would help understand the impacts of monetary policy on quantities in a new technological environment. A higher frequency of

adjustment, as seen in online markets, limits the central banks' ability to affect the economy, making it necessary to induce larger impulses to achieve the same real effects.

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A. Annexes

Annex 1.

Frequency and Duration of Prices: Biweekly and Monthly Window Subsample

Seller	Posted				Regular				Reference			
	Median		Mean		Median		Mean		Median		Mean	
	F (%)	(Días)	F (%)	(Días)	F (%)	(Días)	F (%)	(Días)	F (%)	(Días)	F (%)	(Días)
C	1.69	58.50	0.98	57.88	1.69	58.50	0.98	57.88	1.69	58.50	0.97	57.96
A1	5.00	19.50	9.95	30.20	2.63	37.50	6.72	35.22	0.00	59.00	1.56	46.84
A2	5.41	18.00	10.86	26.97	4.44	22.00	5.41	33.00	0.00	58.00	0.97	50.80
B	0.00	58.00	2.34	49.99	0.00	58.00	1.72	50.72	0.00	58.00	0.45	54.57
Total	3.35	38.75	6.15	40.10	2.16	47.75	3.57	42.97	0.00	58.25	0.97	52.69

Note. The table presents the adjustment frequency and implicit duration for posted, regular, and reference prices. The regular price series are the result of applying the filter (with $j = 14$) described in section 3.1 to the posted price series. The reference price series are derived by applying the methodology in section 3.2 with a monthly mode to the posted price series. To aggregate across products, both the median and mean values of the product prices were used. The adjustment frequency is reported as a percentage, while the implicit duration is reported in days. It is important to note that the upper limit for implicit duration is constrained by the sample size, which is close to 60 days. For this reason, the implicit durations do not exceed 60 days in any case, for example, when the adjustment frequency equals zero

Annex 2.

Size and direction of adjustment in prices biweekly and monthly window sub-sample

Seller	Posted				Regular				Reference			
	Median		Mean		Median		Mean		Median		Mean	
	S (%)	Di (%)	S (%)	Di (%)	S (%)	Di (%)	S (%)	Di (%)	S (%)	Di (%)	S (%)	Di (%)
C	0.28	100.00	1.61	73.89	0.28	100.00	1.61	73.89	0.28	100.00	1.61	73.61
A1	16.25	50.00	17.29	48.17	14.02	50.00	16.34	50.00	12.33	50.00	16.08	49.87
A2	22.31	50.00	22.10	48.23	22.31	50.00	20.74	48.37	15.42	50.00	19.08	47.12
B	19.44	50.00	24.30	53.97	19.43	50.00	24.14	54.30	10.36	50.00	18.20	53.21
Total	17.85	50.00	19.70	51.10	16.72	50.00	18.54	52.15	11.34	50.00	17.14	51.54

Note. The adjustment size is reported as a percentage; however, it is calculated using a logarithmic approximation (see Section 3.4).

Annex 3.

Transition matrix between reference price and transitory price biweekly and monthly window sub-sample.

Seller	State	Reference Price (1)		Transitory Price (2)	
		Median	Mean	Median	Mean
C	Reference Price (1)	98,1%	98,9%	1,9%	1,1%
A1		97,1%	93,9%	2,9%	6,1%
A2		96,7%	93,2%	3,3%	6,8%
B		100,0%	98,5%	0,0%	1,5%
Total		97,4%	94,5%	2,6%	5,5%
C	Transitory Price (2)	20,0%	20,3%	80,0%	79,7%
A1		33,3%	44,3%	66,7%	55,7%
A2		50,0%	55,6%	50,0%	44,4%
B		50,0%	60,3%	50,0%	39,7%
Total		40,0%	49,2%	60,0%	50,8%

Note: To aggregate the values at the store level, the transition matrix was calculated at the product level, and then the mean and median of the values for each store were computed.

Annex 4.

Transition matrix between regular price and transitory price biweekly and monthly window sub-sample.

Seller	State	Regular Price (1)		Transitory Price (2)	
		Mediana	Media	Mediana	Media
C	Regular price (1)	100,0%	100,0%	0,0%	0,0%
A1		100,0%	97,9%	0,0%	2,1%
A2		100,0%	96,7%	0,0%	3,3%
B		100,0%	99,5%	0,0%	0,5%
Total		100,0%	97,8%	0,0%	2,2%
C	Transitory Price (2)				
A1		50,0%	58,9%	50,0%	41,1%
A2		100,0%	83,7%	0,0%	16,3%
B		100,0%	76,2%	0,0%	23,8%
Total		100,0%	72,5%	0,0%	27,5%

Note: To aggregate the values at the store level, the transition matrix was calculated at the product level, and then the mean and median of the values for each store were computed.

Annex 5.

Frequency and duration of prices by purchase categories biweekly and monthly window sub-sample.

Category	Seller	Products	Posted				Regular				Reference		
			Median		Mean		Median		Mean		Median		Me
			F (%)	(Days)	F (%)	d	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)
Alcoholic Beverages, Tobacco And Narcotics	C	31	1.75	56.50	1.63	57.08	1.75	56.50	1.63	57.08	1.75	56.50	1.63
	A1	978	0.00	49.00	4.44	38.92	0.00	59.00	3.26	41.87	0.00	61.00	1.10
	A2	608	16.28	5.63	15.5	19.14	4.76	20.50	6.66	26.33	0.00	58.00	1.35
	B	612	0.00	58.00	1.70	52.21	0.00	58.00	1.13	52.97	0.00	58.00	0.24
	Total	2221	0.00	53.00	6.69	37.43	0.00	56.00	3.58	40.89	0.00	58.00	0.94
Clothing And Footwear	C	-	-	-	-	-	-	-	-	-	-	-	-
	A1	53	0.00	53.00	5.39	42.03	0.00	53.00	4.37	43.31	0.00	59.00	1.28
	A2	13	28.00	3.04	28.0	13.10	23.0	3.81	20.08	13.79	3.85	25.50	3.44
	B	-	-	-	-	-	-	-	-	-	-	-	-
	Total	66	0.00	47.00	9.86	36.33	0.00	47.00	7.47	37.50	0.00	54.00	1.70
Food And Non- Alcoholic Beverages	C	290	1.69	58.50	0.91	58.05	1.69	58.50	0.91	58.05	1.69	58.50	0.91
	A1	1383	10.00	9.49	12.2	20.59	5.00	19.50	6.36	29.64	2.44	40.50	1.82
	A2	2004	5.00	19.50	10.6	27.75	2.63	37.50	5.38	33.89	0.00	58.00	0.73
	B	666	0.00	58.00	2.85	48.65	0.00	58.00	2.18	49.32	0.00	58.00	0.61
	Total	4343	4.76	20.50	9.30	30.70	2.50	39.50	4.90	36.51	0.00	58.00	1.07
Furnishings , Household Equipment And Routine Household	C	-	-	-	-	-	-	-	-	-	-	-	-
	A1	647	4.55	21.50	15.2	28.33	2.86	34.50	13.54	30.72	0.00	55.00	1.96
	A2	356	2.33	42.50	5.33	34.25	2.33	42.50	4.15	36.11	2.22	44.50	1.51
	B	2	3.85	36.75	3.85	36.75	3.85	36.75	3.85	36.75	3.85	36.75	3.85
	Total	999	2.94	33.50	11.73	30.44	2.70	36.50	10.19	32.64	0.00	49.00	1.80
Health	C	1	0.00	60.00	0.00	60.00	0.00	60.00	0.00	60.00	0.00	60.00	0.00
	A1	19	2.70	36.50	11.13	35.43	2.70	36.50	6.47	36.05	0.00	59.00	0.96
	A2	27	22.22	3.98	17.6	11.61	0.00	58.00	1.91	42.68	0.00	58.00	0.08
	B	21	0.00	58.00	1.80	46.16	0.00	58.00	1.80	46.16	0.00	59.00	0.30
	Total	68	5.00	19.50	10.6	29.65	0.00	58.00	3.12	42.16	0.00	58.00	0.40

Categories	Seller	Products	Posted				Regular				Referen		
			Median		Mean		Median		Median		Mean		
			F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	F (%)	(Days)	(Days)
Information And Communication	C	-	-	-	-	-	-	-	-	-	-	-	-
	A1	68	15.89	5.84	29.83	20.25	14.46	6.72	28.73	20.38	2.86	34.50	3
	A2	96	1.11	50.75	4.64	36.41	0.00	58.00	3.14	41.38	0.00	58.00	1
	B	3	0.00	55.00	4.00	40.27	0.00	55.00	1.33	45.83	0.00	55.00	0
	Total	161	5.00	19.50	14.88	29.90	2.63	37.50	13.53	32.91	0.00	55.00	1
Personal Care, Social Protection And Miscellaneous Goods And Services	C	1	0.00	58.00	0.00	58.00	0.00	58.00	0.00	58.00	0.00	58.00	0
	A1	182	2.74	36.00	11.92	31.01	0.00	54.50	6.98	34.37	0.00	55.00	1
	A2	116	9.09	10.49	15.05	15.97	6.82	14.16	7.77	25.00	0.00	56.00	1
	B	31	0.00	52.00	4.53	38.73	0.00	52.00	3.39	40.04	0.00	53.00	1
	Total	330	6.82	14.16	12.29	26.53	4.01	24.75	6.90	31.68	0.00	55.00	1
Recreation, Sport And Culture	C	4	1.72	57.50	1.72	50.87	1.72	57.50	1.72	50.87	1.72	57.50	1
	A1	419	0.00	55.00	3.96	42.63	0.00	59.00	2.38	45.66	0.00	59.00	0
	A2	133	0.00	57.00	5.34	38.93	0.00	57.00	2.48	42.85	0.00	58.00	0
	B	4	0.00	58.50	0.00	56.75	0.00	58.50	0.00	56.75	0.00	58.50	0
	Total	551	0.00	57.00	4.24	41.91	0.00	58.00	2.38	45.11	0.00	58.00	0
Transport	C	-	-	-	-	-	-	-	-	-	-	-	-
	A1	20	0.00	59.00	0.00	58.95	0.00	59.00	0.00	58.95	0.00	59.00	0
	A2	2	5.93	16.91	5.93	16.91	5.93	16.91	5.93	16.91	2.38	41.50	2
	B	-	-	-	-	-	-	-	-	-	-	-	-
	Total	22	0.00	59.00	0.54	55.13	0.00	59.00	0.54	55.13	0.00	59.00	0